Practice workbook · Make an algorithm faster or work for any amount of data

Judging which of two working algorithms is more efficient, or improving a working algorithm so it copes with any amount of data (array instead of separate variables, LENGTH instead of a hard-coded count).

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T1-05

1. Question 4 · part b · 2 marks

2024 JUN · Paper 02

Worked-example retry · Attempt part b only.

4 Programmers write code that keeps track of values, is efficient and uses subprograms.

(a) Global variables are different from local variables.

(i) State where a **global** variable is created.

(1)

(ii) State where a **local** variable is accessible.

(1)

(b) An algorithm reports the range that an integer falls within.

Integers from 0 to 39 are in the low range.

Integers from 40 to 69 are in the middle range.

Integers from 70 to 100 are in the high range.

Figure 4 and **Figure 5** show two algorithms that solve this problem.

```
1  SEND 'Enter an integer: ' TO DISPLAY
2  RECEIVE theNumber FROM (INTEGER) KEYBOARD
3  IF (theNumber < 40) THEN
4      SEND 'Low range' TO DISPLAY
5  ELSE
6      IF (theNumber < 70) THEN
7          SEND 'Middle range' TO DISPLAY
8      ELSE
9          SEND 'High range' TO DISPLAY
10     END IF
11 END IF
```

Figure 4

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8

P 7 5 7 3 7 A 0 8 1 6

1. Question 4 · part b · 2 marks

2024 JUN · Paper 02

Worked-example retry · Attempt part b only.

```
1   SEND 'Enter an integer: ' TO DISPLAY
2   RECEIVE theNumber FROM (INTEGER) KEYBOARD
3   IF (theNumber < 40) THEN
4       SEND 'Low range' TO DISPLAY
5   END IF
6   IF (theNumber > 39) AND (theNumber < 70) THEN
7       SEND 'Middle range' TO DISPLAY
8   END IF
9   IF (theNumber > 69) THEN
10      SEND 'High range' TO DISPLAY
11  END IF
```

Figure 5

Explain the reason that the algorithm in **Figure 4** is more efficient than the algorithm in **Figure 5**.

(2)

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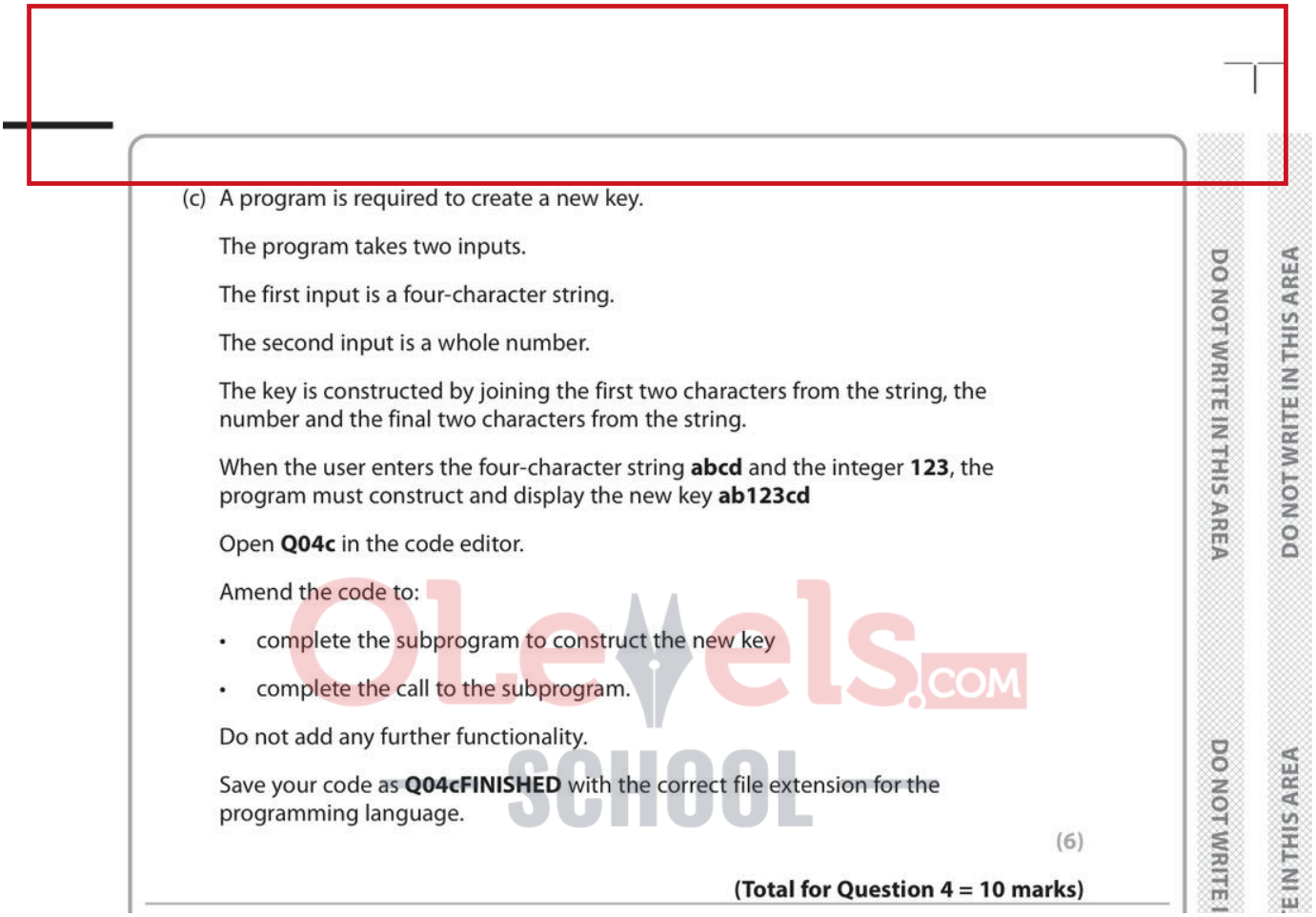
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1. Question 4 · part b · 2 marks

2024 JUN · Paper 02

Worked-example retry · Attempt part b only.



(c) A program is required to create a new key.

The program takes two inputs.

The first input is a four-character string.

The second input is a whole number.

The key is constructed by joining the first two characters from the string, the number and the final two characters from the string.

When the user enters the four-character string **abcd** and the integer **123**, the program must construct and display the new key **ab123cd**

Open **Q04c** in the code editor.

Amend the code to:

- complete the subprogram to construct the new key
- complete the call to the subprogram.

Do not add any further functionality.

Save your code as **Q04cFINISHED** with the correct file extension for the programming language.

(6)

(Total for Question 4 = 10 marks)

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Working and answer

2. Question 6 · part b.ii · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part b.ii only.

6 A software engineer is working with some algorithms.

(a) An algorithm needs to be developed.

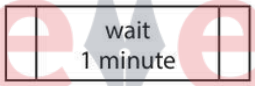
The algorithm must:

- check for a change of hour every minute
- use the library subprogram getHour() to get the hour part of the current time in the 24-hour clock (0 to 23)
- output "Good morning" when the hour is between 3 and 12, inclusive
- output "Good afternoon" when the hour is between 13 and 19, inclusive
- output "Good night" at all other times.

Complete the flowchart to represent this algorithm, in the space provided on the next page.

The call to the library subprogram getHour() has already been included.

Use this subprogram symbol to show waiting for 1 minute.



(6)

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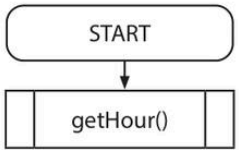
2. Question 6 · part b.ii · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part b.ii only.

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Draw your flowchart here.



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2. Question 6 · part b.ii · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part b.ii only.

(b) Another algorithm determines when to change the flavoured syrups for a drinks dispensing machine.

Figure 4 shows this algorithm written in pseudocode.

```
1  SET flavours TO ["anise", "mango", "cola", "apple", "papaya", "strawberry",
   "lychee", "banana"]
2
3  SET volume TO [0.7, 0.2, 0.6, 0.1, 0.05, 0.8, 0.4, 0.6]
4
5  SET i TO 0
6
7  WHILE i < 8 DO
8
9      IF (volume[i] < 0.1) THEN
10         SEND (flavours[i] & " needs changing") TO DISPLAY
11
12     ELSE
13         IF (volume[i] >= 0.3) AND (volume[i] <= 0.5) THEN
14             SEND (flavours[i] & " needs ordering")  TO DISPLAY
15         END IF
16     END IF
17
18     SET i TO i + 1
19
20 END WHILE
```

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Figure 4

(i) Give the output produced by the algorithm.

(2)

The algorithm works with a fixed number of flavours.

(ii) Give the number of the line in the algorithm that would need to be amended to allow for any number of flavours.

(1)

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2. Question 6 · part b.ii · 1 mark

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Further practice · Attempt part b.ii only.

(iii) State how the pseudocode needs to be changed to make this amendment.

(1)

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QUESTION 6(c) BEGINS ON THE NEXT PAGE.

2. Question 6 · part b.ii · 1 mark · Working and answer

Working and answer

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3. Question 6 · part b.iii · 1 mark

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Further practice · Attempt part b.iii only.

6 A software engineer is working with some algorithms.

(a) An algorithm needs to be developed.

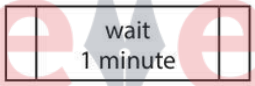
The algorithm must:

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Complete the flowchart to represent this algorithm, in the space provided on the next page.

The call to the library subprogram getHour() has already been included.

Use this subprogram symbol to show waiting for 1 minute.



(6)

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3. Question 6 · part b.iii · 1 mark

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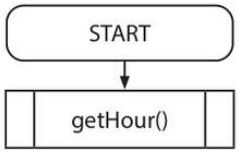
Further practice · Attempt part b.iii only.

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Draw your flowchart here.



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3. Question 6 · part b.iii · 1 mark

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Further practice · Attempt part b.iii only.

(b) Another algorithm determines when to change the flavoured syrups for a drinks dispensing machine.

Figure 4 shows this algorithm written in pseudocode.

```
1  SET flavours TO ["anise", "mango", "cola", "apple", "papaya", "strawberry",
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2
3  SET volume TO [0.7, 0.2, 0.6, 0.1, 0.05, 0.8, 0.4, 0.6]
4
5  SET i TO 0
6
7  WHILE i < 8 DO
8
9      IF (volume[i] < 0.1) THEN
10         SEND (flavours[i] & " needs changing") TO DISPLAY
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13         IF (volume[i] >= 0.3) AND (volume[i] <= 0.5) THEN
14             SEND (flavours[i] & " needs ordering")  TO DISPLAY
15         END IF
16     END IF
17
18     SET i TO i + 1
19
20 END WHILE
```

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Figure 4

(i) Give the output produced by the algorithm.

(2)

The algorithm works with a fixed number of flavours.

(ii) Give the number of the line in the algorithm that would need to be amended to allow for any number of flavours.

(1)

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3. Question 6 · part b.iii · 1 mark

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Further practice · Attempt part b.iii only.

(iii) State how the pseudocode needs to be changed to make this amendment.

(1)

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QUESTION 6(c) BEGINS ON THE NEXT PAGE.

3. Question 6 · part b.iii · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part b.iii only.

(c) The algorithms will be coded in a high-level language.

Programs written in high-level languages must be translated in order to be executed by a computer.

Discuss the similarities and differences between a compiler and an interpreter.

(6)

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(Total for Question 6 = 16 marks)

3. Question 6 · part b.iii · 1 mark · Working and answer

Working and answer

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4. Question 5 · part a · 2 marks

2024 JUN · Paper 02

Further practice · Attempt part a only.

5 Programmers refine algorithms, use searches and read data from files.

(a) **Figure 6** shows an algorithm that determines whether the animal name inputted by the user is found.

```
1  SET a1 TO 'alpaca'
2  SET a2 TO 'bear'
3  SET a3 TO 'camel'
4  SET a4 TO 'deer'
5
6  SEND ('Enter an animal: ') TO DISPLAY
7  RECEIVE target FROM (STRING) KEYBOARD
8
9  IF (a1 = target) THEN
10     SEND ('Found') TO DISPLAY
11 ELSE
12     IF (a2 = target) THEN
13         SEND ('Found') TO DISPLAY
14     ELSE
15         IF (a3 = target) THEN
16             SEND ('Found') TO DISPLAY
17         ELSE
18             IF (a4 = target) THEN
19                 SEND ('Found') TO DISPLAY
20             END IF
21         END IF
22     END IF
23 END IF
```

Figure 6

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DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

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4. Question 5 · part a · 2 marks

2024 JUN · Paper 02

Further practice · Attempt part a only.

Explain **one** improvement to this algorithm that will reduce the number of variables required and will enable it to work with any number of animals.

(2)

(b) Binary search is a divide and conquer algorithm.

Here is a list of data.

10	12	13	14	15	16	17	18	19	20

A binary search is used to check whether the number 11 is in the list.

The midpoint is calculated by adding the high index to the low index and **integer dividing** by 2

Complete the table to show the values for the high index, the low index and the midpoint on each pass of a binary search.

You may not need to use all the rows in the table.

(4)

high index	low index	midpoint

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4. Question 5 · part a · 2 marks · Working and answer

Working and answer

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